

our estimates for transfer pricing that have been developed based upon analyses of appropriate arms-length prices. Similarly, our estimates related to uncertain tax positions concerning research tax credits are based on an assessment of whether our available documentation corroborating the nature of our activities supporting the tax credits will be sufficient. Although we believe that we have adequately reserved for our uncertain tax positions (including net interest and penalties), we can provide no assurance that the final tax outcome of these matters will not be materially different. We make adjustments to these reserves when facts and circumstances change, such as the closing of a tax audit or the refinement of an estimate. To the extent that the final tax outcome of these matters is different than the amounts recorded, such differences will affect the provision for income taxes in the period in which such determination is made and could have a material impact on our financial position and results of operations.

Advertising Expense

Advertising costs are expensed when incurred and are included in marketing and sales expenses in the accompanying consolidated statements of income. We incurred advertising expenses of \$310 million, \$281 million, and \$135 million for the years ended December 31, 2016, 2015, and 2014, respectively.

Cash and Cash Equivalents, and Marketable Securities

Cash and cash equivalents primarily consist of cash on deposit with banks and investments in money market funds with maturities of 90 days or less from the date of purchase.

We hold investments in marketable securities, consisting of U.S. government securities, U.S. government agency securities, and corporate debt securities. We classify our marketable securities as available-for-sale investments in our current assets because they represent investments of cash available for current operations. Our available-for-sale investments are carried at estimated fair value with any unrealized gains and losses, net of taxes, included in accumulated other comprehensive (loss) income in stockholders' equity. Unrealized losses are charged against interest and other income/(expense), net when a decline in fair value is determined to be other-than-temporary. We have not recorded any such impairment charge in the periods presented. We determine realized gains or losses on sale of marketable securities on a specific identification method, and record such gains or losses as interest and other income/(expense), net.

We classify certain restricted cash balances within prepaid expenses and other current assets and other assets on the accompanying consolidated balance sheets based upon the term of the remaining restrictions.

Fair Value of Financial Instruments

We apply fair value accounting for all financial assets and liabilities and non-financial assets and liabilities that are recognized or disclosed at fair value in the financial statements on a recurring basis. We define fair value as the price that would be received from selling an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. When determining the fair value measurements for assets and liabilities, which are required to be recorded at fair value, we consider the principal or most advantageous market in which we would transact and the market-based risk measurements or assumptions that market participants would use in pricing the asset or liability, such as risks inherent in valuation techniques, transfer restrictions and credit risk. Fair value is estimated by applying the following hierarchy, which prioritizes the inputs used to measure fair value into three levels and bases the categorization within the hierarchy upon the lowest level of input that is available and significant to the fair value measurement:

Level 1-Quoted prices in active markets for identical assets or liabilities.

Level 2-Observable inputs other than quoted prices in active markets for identical assets and liabilities, quoted prices for identical or similar assets or liabilities in inactive markets, or other inputs that are observable or can be corroborated by observable market data for substantially the full term of the assets or liabilities.

Level 3-Inputs that are generally unobservable and typically reflect management's estimate of assumptions that market participants would use in pricing the asset or liability.

Our valuation techniques used to measure the fair value of money market funds and marketable debt securities were derived from quoted market prices or alternative pricing sources and models utilizing market observable inputs. Our valuation technique used to measure the fair value of our contingent consideration liability prior to completion of the performance milestones, was based on the present value of probability-weighted future cash flows related to the contingent earn-out criteria and the fair value of our common stock on each reporting date. Upon completion of the performance milestones in the second quarter of 2016, we measure fair value of our contingent consideration liability based on the fair value of our common stock on each reporting date.

Accounts Receivable and Allowance for Doubtful Accounts

Accounts receivable are recorded and carried at the original invoiced amount less an allowance for any potential uncollectible amounts. We make estimates for the allowance for doubtful accounts based upon our assessment of various factors, including historical

Share-based compensation expense included in costs and expenses:

	Three Months Ended							
	Dec 31, 2016	Sep 30, 2016	Jun 30, 2016	Mar 31, 2016	Dec 31, 2015	Sep 30, 2015	Jun 30, 2015	Mar 31, 2015
	(as a percentage of total revenue)							
Cost of revenue	—%	—%	—%	—%	—%	—%	1%	—%
Research and development	7	9	10	11	10	13	15	16
Marketing and sales	1	1	1	2	1	2	2	2
General and administrative	1	1	1	1	1	1	1	1
Total share-based compensation expense	9%	12%	13%	14%	13%	17%	19%	20%

Liquidity and Capital Resources

Year Ended December 31,		
2016	2015	2014
(in millions)		

Consolidated Statements of Cash Flows Data:

Net cash provided by operating activities ⁽¹⁾	\$ 16,108	\$ 10,320	\$ 7,326
Net cash used in investing activities	(11,739)	(9,434)	(5,913)
Net cash used in financing activities ⁽¹⁾	(310)	(139)	(298)
Purchases of property and equipment	(4,491)	(2,523)	(1,831)
Depreciation and amortization	2,342	1,945	1,243
Share-based compensation ⁽¹⁾	3,218	2,960	1,786

(1) We elected to early adopt ASU 2016-09 in the fourth quarter of 2016. The impacts of adoption have been reflected retrospectively in certain of our consolidated statements of cash flows data for all periods presented. Share-based compensation for the year ended December 31, 2016 included the impact the adoption. See Note 1 of the accompanying notes to our consolidated financial statements for additional information related to this adoption.

Our principal sources of liquidity are our cash and cash equivalents, marketable securities, and cash generated from operations. Cash and cash equivalents, and marketable securities consist primarily of cash on deposit with banks, investments in money market funds, and investments in U.S. government securities, U.S. government agency securities, and corporate debt securities. Cash and cash equivalents, and marketable securities were \$29.45 billion as of December 31, 2016, an increase of \$11.02 billion from December 31, 2015, primarily due to \$16.11 billion of cash generated from operations, partially offset by \$4.49 billion for purchases of property and equipment and \$312 million for principal payments on capital lease and other financing obligations.

Cash paid for income taxes (net of refunds) was \$1.21 billion for the year ended December 31, 2016. As of December 31, 2016, our federal net operating loss carryforward was \$3.14 billion, although we anticipate only a relatively small portion of this will be available to offset our federal taxable income in 2017. As of December 31, 2016, we had \$312 million of federal tax credits, of which a substantial portion will be available to offset our federal tax liabilities in 2017. We expect that the amount of cash paid for income taxes will significantly increase in 2017.

In May 2016, we terminated our undrawn five-year senior unsecured revolving credit facility that allowed us to borrow up to \$6.5 billion and entered into a \$2.0 billion senior unsecured revolving credit facility (2016 Facility). Any amounts outstanding under the 2016 Facility will be due and payable on May 20, 2021. As of December 31, 2016, no amounts had been drawn down and we were in compliance with the covenants under this credit facility.

In November 2016, our board of directors authorized a \$6.0 billion share repurchase program of our Class A common stock commencing in 2017 and does not have an expiration date. The timing and actual number of shares repurchased depend on a variety of factors, including price, general business and market conditions, and other investment opportunities, through open market purchases or privately negotiated transactions including through the use of trading plans intended to qualify under Rule 10b5-1 under the Securities Exchange Act of 1934, as amended (Exchange Act).

As of December 31, 2016, \$5.10 billion of the \$29.45 billion in cash and cash equivalents, and marketable securities was held by our foreign subsidiaries. We have provided residual taxes in jurisdictions where we do not intend to indefinitely reinvest the earnings of the local subsidiary.